<u>CSE 555-Final Project</u>

Deadline: 29.05.2026 23:59

DataSet: Find a dataset having more than 10 numerical features and more 2 classes. Briefly explain the dataset you will analyze.

Requested: Using this dataset, you are expected to carry out the items specified below in a Python environment and interpret the outputs obtained after each item. In the report you prepare at the end of the project, you are expected to follow academic writing rules (formatting will be graded). At the end of the project, you are expected to submit both the report and the Jupyter Notebook you used to prepare the project.

1. Plot the distribution of each feature using a boxplot and interpret the results. Apply an outlier detection method of your choice, perform data cleaning, and interpret the results.
2. Analyze the correlation among the features and among the features and class label.
3. Normalize the data using z-score, then calculate the Fisher Distance (between classes) for each feature and analyze the results.
4. Transform the dataset using PCA, treat the projection onto each eigenvector as a feature, and calculate the Fisher Distance for each of these features. Compare these Fisher Distances with those obtained in item 3. Is there a relationship between the eigenvalues and the Fisher Distances of the projected features? Interpret your findings.
5. Plot the projections of the data onto the two most important eigenvectors using a scatter plot and interpret the results. In the scatter plot, display samples from each class in different colors. Repeat the same process for the two least important eigenvectors.
6. Reduce the dataset to 2 dimensions using the LDA algorithm and display it in a scatter plot. In this plot as well, show each class in a different color, and compare the resulting diagram with the one obtained in item 4. Evaluate the performance of the PCA-based and LDA-based diagrams in revealing class separability. (Calculating this with a metric will be considered a bonus.)
7. Cluster the dataset using the K-Means, DBSCAN, t-SNE, and SOM algorithms, and present the results. In <u>**K-Means and DBSCAN**</u>, use the projections onto the first 2 principal components obtained from PCA instead of the original data. <u>**For t-SNE and SOM**</u>, use the original dataset. In these visualizations, display samples from each class in different colors.
8. Select one of the features and propose a hypothesis on the equality of mean in classes and analyze this hypothesis with t-test using 36 and 64 samples randomly drawn from classes.